

The Total-Work Cost of Bounded Retry Guarantees

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An optimistic completion can validate preparation, recompute under protection, or reuse preparation on a match and recompute on a mismatch. Under a call cap, these choices trade total work against protected work. We prove complete polynomial body-work frontiers for independent persistent jobs with arbitrary retry slack, and a least total-work curve for one policy providing optimal protection at every hidden write budget. An all-program domination theorem covers free inspections and retained records. With a supplied one-write bound, no spare calls and free comparisons, charging guards in both resources makes joint caps NP-complete even for independent jobs. With unit guards, independent-job frontiers are computable greedily; joint caps on height-two DAGs are strongly NP-complete despite polynomial output size. Changing how cached recomputation consumes the call allowance also changes the tight protection guarantee. Java wrappers realize the core contract. A native Linux pathname experiment illustrates the code choice and its counting sensitivity; Roslyn preservation and timing results expose transfer limits. The guarantees concern specified operations and resources, without assuming a caller latency requirement.

CCS Concepts: • **Software and its engineering** → **Concurrent programming structures**.

Additional Key Words and Phrases: concurrent programming, retries, resource guarantees, scheduling, online policies

1 Introduction

After optimistic validation fails, a developer can retry cheaply, recompute under protection, or prepare again for guarded reuse and recomputation on mismatch. With bounded validation/completion rounds, which choices meet both total and protected-work limits?

Linux pathname traversal retries under protection. Our candidate prepares again outside, then reuses under protection or rebuilds on mismatch. One write reduces protected events; two writes increase total events without a protection gain (Table 4). Logical/native call units differ. Section 8.2 states the premises; no caller-required cap or latency SLA was found.

With independent works 8 and 16, supplied write bound $B = 1$ and two calls, cache the 8-unit job; on match finish the other fresh, and on mismatch prepare/publish it cheaply. The paths cost (total, protected) work (24, 16) and (32, 8), so the separate worst-case pair is (32, 16). All-fresh gives (24, 24). Requiring one policy with optimal protection at every hidden budget instead forces (40, 16) at $B = 1$. A joint guarantee is a pair of separate maxima, which need not occur on one execution.

We prove polynomial body-work frontiers, an optimal unaware work curve and full-program domination. For supplied $B = 1$, $Q \leq n$ and free comparisons, changing entry charges from zero to one in both coordinates makes DAG joint caps strongly NP-complete despite polynomial frontier size. Independent jobs with common charges retain greedy decision. We also prove call-unit sensitivity and examine transfer in Java, Linux and Roslyn.

2 Completion Interfaces and Resource Contracts

Write $\mathbb{N} = \{0, 1, \dots\}$. Jobs $J = \{1, \dots, n\}$, $n \geq 1$, form a finite DAG with predecessor set $\text{Pred}(i)$. Readiness requires all predecessors completed. Completion records an exact immutable output and persists despite later external writes. The objective is one successful foreground transformation per job, not a simultaneous snapshot or transaction. Policies are deterministic and causal; no randomized expected-cost guarantee is claimed.

Each ready job has a pure, opaque, whole kernel receiving its own-input identity and exact completed predecessor outputs. Every execution performs its fixed work $w_i > 0$, whether inside or outside protection. Partial/incremental evaluation, preparation before readiness and discharge by

external work are excluded in this interface. A preparation is tied to its own-input identity and predecessor tuple. Captured outputs cannot be reconstructed by later reads of the mutable map.

States, histories and turns. A state $(x, \mathcal{M}, \mathcal{R}, \ell)$ contains live own-input identities, completed immutable outputs, a finite snapshot/record store, and program locals. A record carries its job, captured identity, predecessor tuple and computed result. Records may be retained, copied and selected, but not fabricated or retagged without computation. In Table 2, U means no selected record, C a selected possibly stale record, and D permanent completion. A cheap failure clears selection but preserves retained copies. At every finite history each unfinished job admits a legal identity never previously used for that job. Writes preserve completed outputs.

Visible histories list actions, returned values and completion events. Cached completion exposes its identity-comparison Boolean alongside the completed output; pure recomputation may return the same canonical result. Completion publishes its saved output over the live own-key entry. A program chooses a ready job/mode or an unprotected inspection, preparation or local action. The environment then selects a finite word of legal writes: the operation’s *gate*, before its indivisible protected part for a callback. Job/mode is fixed before the gate; later record selection may use only the prior store. A cached mismatch recomputes and completes within the same counted call; its exact comparison has no spurious mismatch. Completing comparison/computation/publication is indivisible to writes, and no guard survives return. Inspection honestly returns the live entry.

Immediate, or just-in-time, preparation captures and computes only for the next selected call. General programs may prepare earlier after readiness, from immutable own snapshots and exact completed parents; old raw snapshots may predate the counted writer interval. Its pure body has no live read or interaction. At completed-operation boundaries, contract it with its returned record and full work, replaying writes since capture before the next operation. Section 6’s relation retains pending writes and does not place a result or full charge at an unfinished concrete prefix.

The common start precedes all computed initialization: every kernel execution contributes to W , including discarded preparations and computation from old raw snapshots. No fixed initialization charge is mandatory. L counts its protected part. Q counts validation/publication/completion calls, including failures; preparations and writers do not consume it. Waiting, common call overhead and nonkernel allocation are excluded. Neither clocks nor work counters are implicitly observed. Write $\Omega_k(T)$ for the sum of the largest $\min(k, |T|)$ works and $\Omega(T)$ for the total; omitted sets mean J , and $\Omega_0 = 0$. These aggregate/allocation objects are inherited [16, 17].

Table 1. Resources and information contracts used in the results.

Symbol	Meaning
$Q \leq n + r$	At most r noncompleting calls beyond the n mandatory completions.
B	Upper bound on external writes; an input only for informed policies.
$a_1 \leq \dots \leq a_n, A_k$	Sorted works and sum of the k smallest works, $A_0 = 0$.
$\mathcal{P}_3(B, r)$	Policies requiring the call cap only against at most B writes.
$\mathcal{U}_3(r)$	One budget-unaware policy requiring the cap at every finite budget.
$\mathcal{V}_r$	Policies in $\mathcal{U}_3(r)$ also attaining optimal L at every budget.

A cheap call publishes only on match and exposes success/failure. Conditional publication and a validation-only callback discarding its raw result implement this transition (Section 6). Two modes offer cheap/fresh operations; three add cached completion. The label “cheap” concerns kernel work. Failed validation leaves the entry unchanged. Free inspections/local record operations neither publish nor retain a guard. Direct acquire/release, retained protection, cross-job callbacks

Table 2. Whole-kernel operations. Every validation or callback adds one to Q ; preparation adds none. Entries show successor state and added (W, L) . Each callback holds protection only within that operation.

Operation	Start	Match	Mismatch
Prepare	U	C, $(w_i, 0)$	—
Cheap publication	C	D, $(0, 0)$	U, $(0, 0)$
Fresh callback	U	D, (w_i, w_i)	—
Cached callback	C	D, $(0, 0)$	D, (w_i, w_i)

and outside completion are excluded, so this is not an arbitrary-lock-API lower bound. Selected operations must terminate. Infinite plays, including endless free actions, have infinite objective.

Strategies and admissibility. An informed strategy is a deterministic function $P(B, h)$; an unaware strategy is $P(h)$. Let $\mathcal{E}_b$ contain gate strategies using at most b writes along every play. For mode family j , $\mathcal{P}_j(B, r)$ consists of strategies completing with $Q \leq n + r$ against every $E \in \mathcal{E}_B$. By contrast, $\mathcal{U}_j(r)$ consists of one B -independent strategy satisfying that cap for every $b \geq 0$ and $E \in \mathcal{E}_b$. The adaptive gate adversary knows the deterministic program and its prior store; it acts after the selected operation, even when a free inspection has just occurred. A fixed deterministic adverse play can be scripted at gates; this does not establish a wall-clock-oblivious or randomized value. Every finite-write play against a fixed program is preserved by capping its environment at that play's write count. Worst cost at budget B is $\sup_{E \in \mathcal{E}_B} L(P, E)$; the two minima use different admissible classes.

LEMMA 1 (FRESH-WRITE AND FINITE-ADVERSARY FACTS). *In Section 2's game, a fresh own-key write at a selected i -operation's gate invalidates every retained i -record. Any concrete prefix has both a no-write and a fresh-write continuation whenever that extra write respects the environment's budget. An adversary bounded by b writes on every play forces any $P \in \mathcal{P}_j(b, r)$ to finish every job, regardless of free inspections or record retention.*

PROOF. A fresh identity differs from every retained target identity. Copying/selection cannot change that, and all observations and job/mode choices agree until the gate. The bounded adversary belongs to $\mathcal{E}_b$; leaving any job unfinished violates admissibility even with infinitely many free actions. Thus every job completes through Table 2. No unbounded adversary is truncated. $\square$

LEMMA 2 (WORK CHARGED BY USEFUL COMPARISON WRITES). *From the charged common start, consider a complete execution in which the adversary writes only fresh own-key identities at comparisons whose zero-write continuation would match. Let H be the multiset of resulting cheap failures and D the set of resulting cached mismatches. Then*

$$W \geq \Omega + \sum_{h \in H} w_{i(h)} + \sum_{i \in D} w_i.$$

Already-stale failures may consume calls, but need no extra charge in this inequality.

PROOF. Charge the computed record selected by the zero-write continuation, even if the actual failure branch selects an older one. The fresh write invalidates every prior target record. Before completion, neither failed cheap calls nor other foreground actions restore an earlier live identity; the restricted adversary never restores one. Each later useful comparison therefore charges a different computed identity. Final completion needs a further distinct execution: protected work or a matching preparation for the final identity. A cached mismatch pays both its pre-gate preparation

and recomputation. Different jobs cannot share charges. This proves the bound with old snapshots, early computation and retained records. The charge is global; a suffix may have prepaid records. $\square$

3 Universal Call and Protected-Work Guarantees

For unfinished jobs S , let $\mathcal{A}(S) = \{i \in S : \text{Pred}(i) \cap S = \emptyset\}$ denote the ready jobs.

THEOREM 1 (PROTECTED WORK UNDER A CALL CAP). *In Section 2's interface, fix integers $B, r \geq 0$. Among programs supplied with B that complete all jobs with $Q \leq n + r$ in every environment with at most B writes, the minimum worst protected work is*

$$L_3^*(B, r) = \Omega_{\min((B-r)_+, n)}, \quad L_2^*(B, r) = \begin{cases} 0, & B \leq r, \\ \Omega, & B > r, \end{cases}$$

with and without cached completion, respectively. Any DAG is permitted.

PROOF. At most r calls may fail to complete. Freshly prepared cheap attempts have disjoint read-to-comparison intervals; each failure needs a distinct write. If $B \leq r$ they give $Q \leq n + B$, $L = 0$. Otherwise switch after r failures to cached completion: at most $B - r$ writes remain, giving the stated bound. All-protected gives the two-mode upper bound.

For two modes when $B > r$, reject cheap calls with fresh writes, stopping unconditionally after $r + 1 \leq B$ rejections. The call cap forbids exhausting this budget, so every job completes under protection.

For three modes, target the $k = \min(B - r, n) > 0$ heaviest jobs. Count noncompleting target comparisons f and completed targets p , initially zero. Write a fresh identity only at target comparison gates while $f \leq r$ and $p < k$; stop permanently at $f = r + 1$ or $p = k$. Counters follow actual outcomes, including fresh completions. While active, cheap target calls fail, cached calls recompute, and fresh completion protects its body without a write.

At operation boundaries the write count satisfies $v \leq f + p$: each write caused a failure or cached completion. Before another write, $v + 1 \leq f + p + 1 \leq r + k \leq B$. This is unconditionally an $\mathcal{E}_B$ strategy, even on plays violating the call cap.

Admissibility excludes $f = r + 1$: those failures plus n mandatory completions violate the cap. Lemma 1 forces every target to complete, fresh or by cached mismatch, protecting its body. Hence $p = k$ and $L \geq \Omega_k$. Free inspections, stale selections and retained copies cannot evade this. Predecessors contribute their own mandatory calls; target predecessors are counted once in p . $\square$

THEOREM 2 (ONE POLICY FOR EVERY WRITER BUDGET). *Let $\mathcal{U}_j(r)$ contain the j -mode programs that receive no B and complete with $Q \leq n + r$ in every finite-write environment. Set $U_j(B, r) = \inf_{P \in \mathcal{U}_j(r)} \sup_{\# \text{writes} \leq B} L(P, E)$. Then*

$$U_3(B, r) = L_3^*(B, r), \quad U_2(B, r) = \Omega \mathbf{1}[B \geq r].$$

One policy per mode family attains these values simultaneously for all B and positive works on any DAG, using only readiness and a global cheap-failure counter.

PROOF. Switch after r cheap failures to a fresh read/preparation with cached completion (three) or to fresh protected completion (two). This uses at most r extra calls universally. At $B < r$ no protection is needed; at $B = r$ cached completions after switching still match. Theorem 1 supplies the remaining upper and three-mode lower bounds.

For two modes when $B \geq r$, reject every cheap call with a fresh own-key write until r rejections, then stop. Only protected calls complete before this. Afterwards, another rejectable call could fail in a different finite environment, violating the call cap. Thus all jobs complete under protection with

at most $r \leq B$ writes. That additional hypothetical write is absent from the lower-bound execution. Retained preparations and inspections cannot prevent a fresh write before comparison. $\square$

For equal works and $B > r$, the protection ratio is $n/\min(B - r, n)$. At $B = r$, informed two-mode policies know no write remains; unaware policies must tolerate another-write extensions. Per-update and shared-batch call allowances are distinct; the Roslyn comparison reports both caps.

4 The Total Work Needed for Optimal Protection

The protection optimum alone permits wasteful preparation. We minimize total work under simultaneous and supplied-budget protection contracts, for arbitrary r . The separate suprema ($\sup W, \sup L$) may have different witnesses; they do not determine $\sup(W + \lambda L)$. All computed initialization remains charged.

4.1 One Policy with Optimal Curves at Every Budget

Let $\mathcal{V}_r$ contain the policies in $\mathcal{U}_3(r)$ satisfying $L(P, E) \leq \Omega_{(b-r)_+}$ for every $b \geq 0$ and $E \in \mathcal{E}_b$. This requires one *simultaneous optimal-protection curve*; it is stronger than universal completion. Write $W_P(B) = \sup_{E \in \mathcal{E}_B} W(P, E)$.

LEMMA 3 (CAPPING A REALIZED PLAY). *For $P \in \mathcal{V}_r$, a finite prefix containing d actual writes has protected work at most $\Omega_{(d-r)_+}$.*

PROOF. Continue the prefix without writes. Admissibility forces completion. Cap that environment unconditionally at d : induction preserves each action and realized write, and there are no later writes to remove. The resulting environment belongs to $\mathcal{E}_d$ and preserves the complete play. Its total protected work satisfies the bound; nonnegative costs give the prefix bound. The program need not observe d . $\square$

THEOREM 3 (LEAST TOTAL-WORK CURVE UNDER SIMULTANEOUS OPTIMAL PROTECTION). *For independent jobs with sorted positive works $a_1 \leq \dots \leq a_n$,*

$$\inf_{P \in \mathcal{V}_r} W_P(B) = F_r(B) := \begin{cases} \Omega + Ba_n, & B \leq r, \\ \Omega + \max_{1 \leq m \leq \min(B-r, n)} \{ra_{n-m+1} + \Omega_m\}, & B > r. \end{cases}$$

One policy attains this entire curve: process jobs in ascending work order, freshly prepare before each cheap call until r cheap failures, and freshly prepare before each cached completion thereafter. It observes neither B nor cached comparison results. Sorting, prefix sums and prefix maxima construct the curve in $O(n \log n)$ arithmetic/comparison operations, representing its initial linear segment analytically.

PROOF. *Upper bound.* Each bad comparison needs a write in its distinct preparation-to-comparison interval. There are at most r cheap failures; cached mismatches occur only thereafter, at distinct jobs. Thus $Q \leq n + r$ and $L \leq \Omega_{(B-r)_+}$ at every budget. When $B \leq r$, at most B preparations are lost, each costing at most a_n .

For $B > r$, suppose cached mode starts at rank j and has $m \geq 1$ mismatches. All failed preparations were at ranks at most j and the m mismatched jobs lie in the suffix, so $j \leq n - m + 1$. Duplicate work is at most $ra_j + \Omega_m \leq ra_{n-m+1} + \Omega_m$, with $m \leq B - r$. Without a mismatch it is at most ra_n , below the $m = 1$ term. For each permitted m , match the first $n - m$ jobs, fail cheap r times at the next job, and mismatch all m cached completions. This attains its term using $r + m$ writes. The analogous maximum-job witness attains $\Omega + Ba_n$ at $B \leq r$.

Lower bound for arbitrary programs. Fix $m \leq \min(B - r, n)$ and target the m heaviest jobs. Write a fresh own-key identity only at target comparisons whose zero-write continuation would match. Let f count these useful cheap failures, c useful cached mismatches and t all target completions.

Stop forever after $f = r + 1$ or $t = m$. At operation boundaries the write count is $d = f + c$, with $c \leq t$. Before another write, $d + 1 \leq f + t + 1 \leq r + m$. This is a globally $(r + m)$ -bounded adversary even on inadmissible programs.

An admissible program cannot reach $f = r + 1$, and every target eventually completes. Before all targets finish, no target comparison can match: a useful comparison is invalidated, and a stale one already fails or protects. Thus all targets complete protectively and $L \geq \Omega_m$. Lemma 3, applied at the realized final count d , gives

$$\Omega_m \leq L \leq \Omega_{(d-r)_+}, \quad d = f + c \leq r + m.$$

Positivity forces $d = r + m$, hence $f = r$ and $c = m$. All target completions are useful cached mismatches; this is a consequence of the contract, not an assumed normal form. Lemma 2 now charges r distinct failed preparations of work at least a_{n-m+1} , all m duplicate target kernels and mandatory Ω . Maximizing over m gives the lower bound.

Early budgets. The $m = 1$ witness just constructed has r useful cheap failures at a maximum-work job, then its useful cached mismatch. Truncate after the first $B \leq r$ writes. Its unchanged prefix has destroyed B distinct paid preparations of that unfinished job. Eventual completion needs another distinct kernel, giving $\Omega + Ba_n$ by global charging. At $B = 0$ mandatory work suffices. Lower bounds include retained records, free inspections and paid early preparations after readiness. $\square$

The maximizing m need not be largest: works $(1, 100)$ and $r = 1$ give terms 200 and 102. Optimal W and optimal L may have different maximizing executions. The theorem identifies one least attainable pair of curves, not the work of every optimal- L policy. A relaxed protection requirement defines a different comparison.

Dependency boundary. The lower bound is order-independent. For any DAG, any topological cheap-then-cached policy attains $F_r(B) = \Omega + Bw_{\max}$ at $B \leq r + 1$, while retaining optimal L at all budgets. At $r = 0$, all-cached gives $F_0(B) = \Omega + \Omega_B$ for every DAG and budget. Later positive-retry values can differ: the chain $10 \rightarrow 1 \rightarrow 1$, $r = 1$, $B = 3$, requires $W = 33$, versus 32 for independent jobs. Two target writes force a cheap failure and cached mismatch at the 10-unit job. Its protected work already equals $\Omega_1 = 10$; the next job cannot be fresh on the no-more-write continuation, nor cheap with no retry left. A third write duplicates that unit: mandatory work 12, a failed 10-unit preparation, a cached 10-unit duplicate and one unit give $12 + 10 + 10 + 1 = 33$. The chain policy attains this value. A least entire curve for arbitrary positive-retry DAGs remains unresolved.

4.2 The Complete Frontier for Independent Jobs

A supplied write bound permits a different policy for each resource contract. Here allow arbitrary retry slack $r \geq 0$. Sort independent jobs by positive work $a_1 \leq \dots \leq a_n$ and let $A_k = \sum_{j=1}^k a_j$, $A_0 = 0$. Put $h = \min(r, B)$ and $p = B - h = (B - r)_+$.

THEOREM 4 (INDEPENDENT-JOB FRONTIER). *For independent jobs, the exact $(\sup W, \sup L)$ frontier over $\mathcal{P}_3(B, r)$ is the nondominated subset of (Ω, Ω) and*

$$(\Omega + C_k, \Omega - A_k), \quad 1 \leq k \leq n - p,$$

$$C_k = ha_k + A_{k+p-1} - A_{k-1}.$$

For $p = 0$ the prefix difference is zero. If $p \geq n$ only (Ω, Ω) remains; at $B = 0$ the frontier is $(\Omega, 0)$. Sorting and a prefix-sum sweep construct the frontier in $O(n \log n)$ arithmetic/comparison operations.

PROOF. For selected k , process jobs in ascending work order. Count matching completions s and cheap failures f . Before each preparation, if $s = k$, finish the suffix fresh; otherwise prepare/cheap

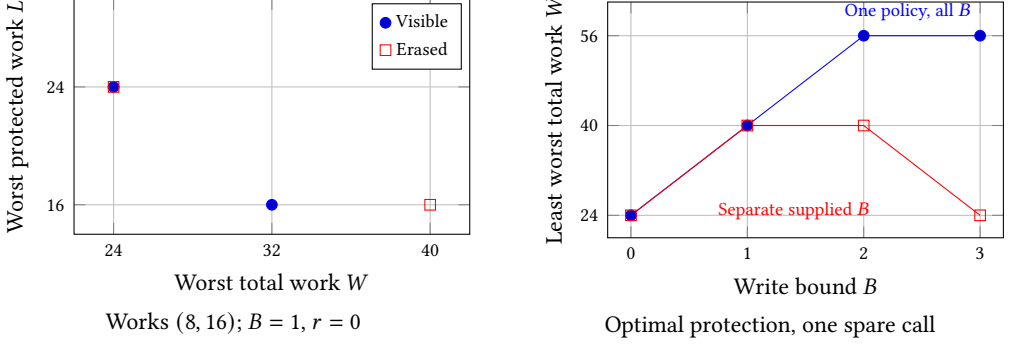


Fig. 1. Independent works (8, 16), counting kernel work. Left: exact visible/erased frontier points, without interpolation. Right: both contracts attain $L^*(B) = (0, 0, 16, 24)$ at $B = 0, \dots, 3$. One unaware policy meets all bounds; supplied policies may differ. Their total optimum can fall as the protection requirement loosens.

when $f < r$ and prepare/cached when $f = r$. Stop when all jobs finish. This policy reads no B after k is selected.

Let m count cached mismatches. Disjoint invalidation intervals give $f + m \leq B$; cached mode requires $f = r$, so $m \leq p$. If all jobs finished before $s = k$, then $n = s + m \leq k - 1 + p < n$, a contradiction. Hence exactly k matching jobs precede the fresh suffix. Their weight is at least A_k , giving $L \leq \Omega - A_k$, attained without writes.

Cheap failures occur before cached mode, at rank $s + 1 \leq k$, and cost at most ha_k . The j th cached mismatch has rank $s + j \leq k + j - 1$. Thus $W - \Omega \leq ha_k + \sum_{j=k}^{k+p-1} a_j = C_k$. To attain it, match the first $k - 1$ jobs, fail cheap h times on job k , then cause p cached mismatches at ranks $k, \dots, k + p - 1$. The next job $k + p \leq n$ matches and triggers fresh completion. For $p = 0$, job k itself matches after the failures. The call bound is $Q \leq n + h$ within budget and $Q \leq n + r$ against any finite number of writes.

For the lower bound, target ranks $k, \dots, n$. A globally B -bounded adversary writes only at target comparisons whose zero-write continuation would match. Already-stale cheap calls receive no write; stale cached calls protect without one. If the budget is not exhausted, every target completes protectively, giving $L \geq \Omega - A_{k-1}$. Otherwise let $f \leq h$ writes cause cheap failures; the other $B - f$ cause cached mismatches at distinct targets. Lemma 2 gives

$$W - \Omega \geq fa_k + \sum_{j=k}^{k+B-f-1} a_j \geq C_k.$$

The actual distinct targets ensure this window exists. Increasing f replaces its last weight by a_k and cannot increase it; at $f = h$ it equals C_k . Wasted stale failures consume retry allowance without reducing this bound.

Consequently worst $L < \Omega - A_{k-1}$ forces worst $W \geq \Omega + C_k$. Since C_k is nondecreasing, a cap $M < C_1$ forces all work protected. Otherwise choose the largest k with $C_k \leq M$; the next index's target bound forces $L \geq \Omega - A_k$. At $k = n - p$, Theorem 1 supplies the same lower bound Ω_p . The attaining policies complete the frontier; nondominance removes tied thresholds. At $B = 0$, select $k = n$; at $p \geq n$, mandatory work and the protected-work lower bound give all-fresh. $\square$

Selecting k uses the supplied contract. Its constant protection cap holds exactly through $B = r + n - k$; the next budget permits r failures and $n - k + 1$ cached mismatches.

4.3 When Retry Slack Covers the Write Budget

PROPOSITION 5 (AN ORDER-INDEPENDENT FRONTIER WITHOUT THE CACHED FLAG). *If $r \geq B > 0$, on every DAG the least worst L under $W \leq \Omega + M$ is*

$$\sum_{i: B w_i > M} w_i.$$

Its attaining policy needs only cheap success/failure, even under the output-only cached contract of Section 5.2.

PROOF. Choose the k smallest jobs for cheap calls and all others fresh; follow any topological order until B cheap failures, then prepare/cache every unfinished job. Duplicate work is at most Ba_k , and protected work at most $\Omega - A_k$. Repeated writes at a maximum chosen job and the no-write path attain the separate bounds. Theorem 4's target lower bound uses no execution order and specializes to $C_k = Ba_k$. Tied thresholds give the displayed cap form. $\square$

4.4 If a Cached Mismatch Consumes Another Call Unit

A logical completion call need not equal a consumer's resource unit. Keep the same atomic operations and observations, but charge a cached mismatch two units; every other cheap/fresh/cached call costs one. Let Q' count these units. At a boundary with u completions and z cheap failures plus cached mismatches, $Q' = u + z$; at completion it is $n + z$. This changes accounting, without splitting the callback into interleavable operations.

PROPOSITION 6 (DEPENDENCE ON THE CALL UNIT). *Under $Q' \leq n + r$, the supplied-budget body frontier on every DAG is $\{(\Omega, \Omega)\}$ when $B > r$, $\{(\Omega, 0)\}$ when $B = 0$, and Proposition 5's frontier when $0 < B \leq r$. Among policies admissible at every finite budget, one policy attains the least protected-work curve 0 for $B < r$ and Ω for $B \geq r$.*

PROOF. For $B > r$, write a fresh own-key identity at every cheap/cached gate whose no-write continuation would match. Stop permanently at $z = r + 1$. At boundaries $v \leq z$, so each useful write while $z \leq r$ respects $v + 1 \leq r + 1 \leq B$, even on inadmissible histories. Every cheap call fails and every cached completion mismatches while active. Admissibility forbids $z = r + 1$; finite completion therefore protects every body. All-fresh attains (Ω, Ω) .

For $B \leq r$, $Q' \geq Q$ inherits the original lower bound. Proposition 5's attaining policy has at most B cheap failures and only matching cached calls after budget exhaustion, so $Q' = Q$. All cheap gives the $B = 0$ pair.

For the unaware upper bound, use cheap calls with immediate preparation until r failures, then all-fresh. For its lower bound at $B \geq r$, use the same reject-all-comparisons adversary but stop at $z = r$, using at most r actual writes. Every earlier completion is protected. If unfinished jobs remain, any further cheap/cached call is already stale or could become bad after one more write in another finite environment, violating universal admissibility. That hypothetical write is absent from the realized play. Thus all remaining completions are fresh. If every job completed earlier the bound already holds. Lemma 1 handles retained records, free inspections and mandatory finite completion. $\square$

Section 8.2 compares logical completion blocks with native call counts.

5 A Complete Finite Policy Class for Arbitrary Retry Slack

We now justify finite policy search for general DAG/budget combinations against all admitted programs.

THEOREM 7 (RESOURCE DOMINATION BY IMMEDIATE-PREPARATION TREES). *For every $P \in \mathcal{P}_3(B, r)$ there is a finite fresh/cheap/cached tree T such that*

$$\forall E \in \mathcal{E}_B \exists E' \in \mathcal{E}_B : (Q, W, L)(T, E) \preceq (Q, W, L)(P, E').$$

Its depth is at most $n + r$, and it prepares only immediately before its selected cheap/cached node.

PROOF. *Extraction.* Restrict the adversary to no writes except an optional fresh own-key write at a comparison whose zero-write continuation matches. Fix a concrete fresh identity at each branch history. A fresh call gives a unary fresh node. A matching cheap call gives completing and fresh-write-failing children; a matching cached call gives two completing children. An already-stale cached call becomes unary fresh, retaining its original zero-write continuation. Delete an already-stale cheap call and retain its failing continuation and fixed local choices in the witness history.

Finite call structure. If the original prefix has u completions, h useful cheap failures and s deleted stale failures, its calls number $u + h + s$; the tree has $u + h$. Its unused failure allowance may increase, but need not be spent. All represented paths have at most $n + r$ nodes. Between them, no-more-write computation and deleted calls must terminate, or the original program would not complete in that finite-write environment. Finite branching and bounded depth therefore yield a finite tree.

Cap the restricted adversary at B and replace a zero-remaining-budget suffix by ready prepare/cached completions. Useful comparisons cost counted calls; the original cap forces completion along every represented path.

Global charging. On a complete path let H be its binary cheap failures and D its binary cached-mismatch jobs. Lemma 2 lower-bounds original work by $\Omega + \sum_H w_i + \sum_D w_i$. Original protected work is at least the sum of the extracted fresh-node works and $\sum_D w_i$; a collapsed stale cached call already protects its kernel. A known-zero-budget suffix substitution can only lower protected work. Deleted stale cheap calls need no separately assigned preparation charge.

Realization and domination. Realize the fixed tree using actual current inputs and exact saved parents, ignoring their values for scheduling. Prepare only at its selected cheap/cached node, never speculatively for a fresh node. Readiness follows the same completed set. Actual failures/mismatches have disjoint preparation-to-comparison intervals and inject into distinct actual writes; irrelevant or identity-reusing writes may remain invisible. Thus a path with $d = |H| + |D|$ bad comparisons requires at least d writes. Exactly d own-gate fresh writes realize that path and its original concrete witness. Define the witness adversary with an unconditional cap d , stopping after that many writes on every history. It is globally d -bounded, rather than merely spending d on one play.

The tree's exact path costs are $Q = n + |H|$, $W = \Omega + \sum_H w_i + \sum_D w_i$, and $L = \sum_{\text{fresh}} w_i + \sum_D w_i$. They are no greater than the original witness's. Any actual tree execution with at most B writes is therefore matched to an original environment in $\mathcal{E}_d \subseteq \mathcal{E}_B$. Correct outputs follow from the operations' semantics, not from reproducing the original program's concrete values under the same environment. $\square$

5.1 Exact Pareto Recurrence

Let $\mathcal{F}(S, b, a)$ contain nondominated suffix-tree bounds (e, l) , where S is unfinished, b is a conservative remaining write allowance, a remaining noncompletions, $e = \max W - \Omega(S)$ and $l = \max L$. Set

$$\mathcal{F}(\emptyset, b, a) = \mathcal{F}(S, 0, a) = \{(0, 0)\}.$$

For ready i , write $R = S \setminus \{i\}$ and take the nondominated union of

$$\begin{aligned} \text{fresh:} & \quad (e, w_i + l), \quad (e, l) \in \mathcal{F}(R, b, a); \\ \text{cached:} & \quad (\max\{e_0, w_i + e_1\}, \max\{l_0, w_i + l_1\}); \\ \text{cheap, } a > 0 : & \quad (\max\{e_0, w_i + e_1\}, \max\{l_0, l_1\}). \end{aligned}$$

Both comparisons choose their matching child in $\mathcal{F}(R, b, a)$. Cached chooses its other child in $\mathcal{F}(R, b - 1, a)$; cheap chooses it in $\mathcal{F}(S, b - 1, a - 1)$. Children are selected independently.

LEMMA 4 (SUFFICIENT STATE FOR A SUFFIX TREE). *The recurrence gives exactly the nondominated immediate-preparation tree bounds. Here b counts the remaining allowed bad branches; it need not equal actual unobserved writes remaining.*

PROOF. A path with d bad branches needs at least d distinct invalidating writes. Conversely, exactly one fresh own-key write at each bad gate and no others realizes that path. Hence worst costs with at most b writes equal maxima over paths with at most b bad branches. Inert writes or multiple writes at one gate add no new path.

Induct on $|S| + a$. At $S = \emptyset$ there is no cost. At $b = 0$, prepare/cached completes every ready job with no duplicate or protected work, dominating other suffixes. Otherwise a fresh node has one child on R . A comparison has a matching child with allowance b and a bad child with allowance $b - 1$; a cheap failure additionally consumes one of a retries and leaves S unchanged. Each child decreases the induction rank. Readiness depends only on S ; under immediate preparation, costs depend only on the selected job/mode and outcome. These facts yield exactly the displayed charges.

The two children describe mutually exclusive observations. The controller may choose either continuation independently in advance; the adversary follows only one. It therefore receives no split budget across children. Taking a maximum separately in each resource is both necessary and sufficient for the two worst-case caps. Conversely every tree has precisely this root/child decomposition. Monotonicity of each charge/max expression allows a dominated child to be replaced by a dominating one. Emitted policies retain witness nodes; original histories with the same (S, b, a) need not be equivalent. Theorem 7 supplies full-program completeness. $\square$

There are at most $(r + 1)(\min(B, n + r) + 1)2^n$ states, with possibly exponential frontier sizes. The independent-job and enough-retry DAG formulas identify polynomial regions; a general fixed-order theorem is not claimed.

The least unaware work curve for general DAGs remains open.

5.2 What the Cached Comparison Reveals

Consider an erased observation contract. Every invocation returns the same immutable per-job object z_i , which completion publishes over its live entry. The cached primitive's *entire visible transition* exposes only that output and completion: no compared input, Boolean, comparison-dependent record selection or callback-local side effect survives. Kernel internals, work counters, allocation IDs and clocks are hidden. Free outside inspections and old snapshots remain allowed. Constant output alone is insufficient if the callback saves the current input elsewhere.

PROPOSITION 8 (A STRICT OBSERVATION BOUNDARY). *For independent works $(8, 16)$, supplied $B = 1$ and $Q \leq 2$, the visible-comparison frontier is $\{(24, 24), (32, 16)\}$; erasure changes it to $\{(24, 24), (40, 16)\}$.*

PROOF. All-fresh attains $(24, 24)$. For $(32, 16)$, cache the 8-unit job; on match complete the other fresh, and on mismatch prepare/cache it. The latter branch certifies the sole write. Theorem 4 gives the visible frontier.

For erasure, follow any admitted program's no-write run. Cheap would permit an extra failed call, so both completions are guarded. Partition them into matching cached C and fresh/already-stale cached F . If F is nonempty, either both bodies are already protected or a single own-gate write at the remaining matching cached job protects its body too. Erasure restores the same visible state after that completion, preserving subsequent inspections, choices and protected F bodies. Thus worst $L \geq 24$ is unavoidable.

If $L < 24$, both jobs must instead be matching cached on that path. One fresh write at the 16-unit job forces $W \geq 24 + 16 = 40$ and $L \geq 16$, by Lemma 2. Immediate prepare/cache for both attains $(40, 16)$: at most one comparison can mismatch. These cases exhaust programs and prove the erased frontier. $\square$

A complete erased-observation frontier for general positive-retry DAGs remains open.

6 Actual External Writes in Java

Use Java 17 ConcurrentHashMap, one controller, permanent keys, deeply immutable non-null values, identity equality and never-reused own-key write identities. Kernels read captured own inputs and saved completed parents; kernels/writers preserve tags/shapes. Key methods are stable; methods terminate without side effects or recursive updates. Replacement may call equals [22, 23]. Normal progress is a premise.

Cheap wraps replace or validation-only compute, exposing success/failure. Cached exposes its saved output and callback-local comparison Boolean; fresh always computes. Observations retain opaque-reference equality, payloads, wrapper results, saved parents and selector state. Numeric IDs, clocks, counters, raw cheap-mismatch values and diagnostics are hidden.

Correspondence. An injective reference map η preserves aliases across live entries, pending writes, saved results and retained handles. Set $\mathcal{M}(i) = \eta(\text{done}[i])$ from the saved output. The store maps exactly available records to $(i, \eta(s), \mathcal{M}|_{\text{Pred}(i)}, \eta(z))$; discarded results are unavailable. Locals map to ℓ .

Let Δ queue committed external assignments deferred in normalization. At boundaries,

$$\text{decode}_\eta(H.\text{map}) = \text{apply}(\Delta, x), \quad b_H = b_x + |\Delta|.$$

Here b counts publications, not diagnostic increments. Decoding follows forwarding nodes; Δ replays assignments, not callbacks. Atomic compute invokes its callback once [21]; the pinned implementation serializes same-key updates in list/tree bins [23]. Wrappers return saved operation results even if the live entry is newer. Completion saves that output, never a reread.

THEOREM 9 (RESOURCE BOUNDS FOR ACTUAL WRITES). *Under these assumptions, the call/body-work results of Sections 3–5 hold for actual external writes, under each result's observation contract and with separate or shared bins. Completed-operation traces preserve results and Q, W, L , and hence Q' under Section 4.4. This is not a mechanically verified JDK/JMM theorem.*

PROOF. Group each completed pure kernel with its operands, result and work. Defer writes since capture in Δ and flush before the next operation, including inspection. Unfinished instructions are not zero-cost stutters. Post-publication/pre-return writes also enter Δ : saved outputs fix $\mathcal{M}$ while the map may be newer. Flush terminal queues without adding writes. Resize/tree conversion copies references; concurrent publications remain writes.

Upper policies associate bad comparisons with distinct committed writes through disjoint fresh preparation-to-comparison intervals. Progress transfers their bounds. For lower bounds, suspend after job/mode/store selection, before API entry. A terminating own-key writer installs a fresh reference without a foreground monitor held, even with colliding bins, invalidating all retained

target records. Bounded target adversaries preserve write caps, mandatory completions and global charges. Further finite continuations establish universal admissibility. Saved parents preserve readiness; callback Booleans implement observable branches. $\square$

Earlier kernels perform $8w_i$ counted iterations; newer wrappers use w_i units. Waiting, allocation, traversal and writers are outside W, L . Counters are checker-only; the finite matrix excludes concurrent resize.

7 Charging Entry Work Changes Joint-Cap Complexity

Body-work accounting isolates kernel recomputation. A protected callback also enters a guard and may compare a record. Give these actions nonnegative supplied costs g_i and c_i , respectively; fresh completion pays only g_i , and cheap calls have zero added charge. Define

$$W_t = W + \sum_{\text{guarded}} g_i + \sum_{\text{cached}} c_i, \quad P_t = L + \sum_{\text{guarded}} g_i + \sum_{\text{cached}} c_i.$$

These mathematical prices are not calibrated native times. The guard ends on return; charges add no observations, skipped comparisons or cross-job amortization. Cached mismatch still completes in one call. Throughout this section, $B = 1$ is supplied and $Q \leq n$. A protected call fee $d_i \geq 0$ common to all three modes adds $\sum_i d_i$ to both coordinates: every job completes once and $Q = n$. The formulas omit this policy-independent constant.

For a topological order π and cached subset C , set $F = J \setminus C$, $h_i = g_i + c_i \mathbf{1}[i \in C]$, and $H = \sum_i h_i$. Let H_i sum h_j through i and F_i sum the works of F jobs strictly before i . Empty maxima are 0.

THEOREM 10 (COMPLETE ONE-WRITE REPRESENTATION). *The complete frontier is the nondominated set over all (π, C) of*

$$\begin{aligned} W(\pi, C) &= \Omega + \max\{H, \max_{i \in C} (w_i + H_i)\}, \\ P(\pi, C) &= \max\{H + \Omega(F), \max_{i \in C} (w_i + H_i + F_i)\}. \end{aligned}$$

An order, a mode bit per job and a first-mismatch flag specify an attaining policy.

PROOF. Follow π , using fresh on F and immediate prepare/cache on C until the first mismatch; then prepare/cheap-complete the suffix. A mismatch of this just-prepared record certifies the sole write, so all suffix calls succeed. Without a mismatch, costs are $(\Omega + H, H + \Omega(F))$. A first mismatch at i gives $(\Omega + w_i + H_i, w_i + H_i + F_i)$. No writes and one selected own-gate write realize these cases.

For an arbitrary admitted program, fix a no-write execution. Cheap is impossible: a fresh gate write would add a failure to n mandatory completions. Guarded completions induce π ; take matching cached jobs as C and fresh/already-stale cached jobs as F . Dropping stale cached comparison/preparation charges only weakens the lower bound. This execution costs at least $(\Omega + H, H + \Omega(F))$.

For each $i \in C$, preserve the prefix, write one fresh own-key identity at its gate, then stop. This invalidates all retained target records. Lemma 2 forces body work at least $\Omega + w_i$, including paid early preparation. The incurred guarded/comparison prefix is at least H_i ; protected bodies include $F_i + w_i$. Thus this alternative costs at least $(\Omega + w_i + H_i, w_i + H_i + F_i)$. Each environment has an unconditional one-write cap. The program's separate suprema dominate every witness and hence the displayed pair. The attaining policy proves completeness, including input-dependent choices and retained records. $\square$

7.1 Free Comparisons: Hard Joint Caps, Tractable Special Cases

Now let $c_i = 0$ and $\Gamma = \sum_i g_i$. For independent jobs, move every cached job before every fresh job and sort the cached jobs by nonincreasing w_i . Define

$$K(C) = \max_{i \in C} \left(\sum_{j \in C \text{ through } i} g_j + w_i \right), \quad K(\emptyset) = 0.$$

LEMMA 5 (ONE CANONICAL PAIR PER SUBSET). *The independent-job frontier is the nondominated set of*

$$A(C) = \Omega + \max\{\Gamma, K(C)\}, \quad D(C) = \max\{\Gamma + \Omega(J \setminus C), K(C)\}.$$

PROOF. Exchanging adjacent fresh a then cached b removes g_a and w_a from b 's respective prefixes; all other cached prefixes are unchanged. Both costs in Theorem 10 weakly decrease. Once cached jobs come first, only $K(C)$ depends on their order. For adjacent cached a, b with $w_a < w_b$ and earlier guard sum t , the old maximum is $t + g_a + g_b + w_b$. Both terms after swapping are no larger. Independence permits these exchanges. The theorem now gives the displayed pair and completeness. $\square$

For binary integer works, nonnegative integer guards and integer caps K_W, K_P , set $M = K_W - \Omega$ and $T = \min(M, K_P)$. Reject if $\Gamma > M$ or $\Gamma > K_P$. Otherwise feasibility is exactly the existence of a descending-work subset with

$$p + g_i + w_i \leq T \text{ at each selected job,} \quad \sum_{i \in C} w_i \geq \Gamma + \Omega - K_P,$$

where p is its previously selected guard sum. This is weighted on-time subset selection with processing g_i , profit w_i and deadline $T - w_i$: earliest-deadline order is descending work. Lawler-Moore's inherited dynamic program [2, 3] stores, for each p , the largest saved work. Skipping preserves a state; choosing adds (g_i, w_i) when the prefix test passes. A new layer per job also handles $g_i = 0$. This gives $O(n(\Gamma + 1))$ arithmetic operations and $O(\Gamma + 1)$ decision space. The deadline/profit linkage means unrestricted scheduling hardness alone does not prove hardness here.

THEOREM 11 (COMPARISON-FREE JOINT-CAP BOUNDARY). *Joint-cap decision is NP-complete even for independent jobs with $c_i = 0$ and positive integer g_i, w_i . This class has the preceding pseudopolynomial algorithm. For independent jobs with all $g_i = g$, decision takes $O(n \log n)$ and its complete frontier takes $O(n^3)$ arithmetic operations.*

PROOF. An order/mask certifies the caps. Reduce positive SUBSET SUM $a_1, \dots, a_m$ with $1 \leq a \leq S = \sum_i a_i$ [4]. Make $2m$ independent jobs with $b = (a_1, \dots, a_m, 0, \dots, 0)$, and set

$$d = q = S + 1, \quad g_i = q(d + b_i), \quad \Gamma = q(2md + S), \quad k = \Gamma + S + 1, \quad w_i = k + b_i.$$

Let $\ell = md + a$, $M = k + q\ell + S$, $K_W = \Omega + M$, and $K_P = \Gamma + mk + S - a$. All integers have polynomial binary length. We have $\Gamma < M < K_P$, since

$$K_P - M = q(md + S - a) + (m - 1)k - a > 0.$$

Thus $T = M$. The constraint $K(C) \leq M$ is equivalent to $\sum_C g_i \leq q\ell$: this guard bound makes every prefix plus w_i at most M ; otherwise guard-sum granularity and the last selected job give $K(C) \geq q(\ell + 1) + k > M$. It follows that $|C|d + \sum_C b_i \leq md + a$, so $|C| \leq m$. The protected cap requires $\sum_C w_i \geq mk + a$; because $k > S$, it forces $|C| \geq m$. Hence $|C| = m$ and $\sum_C b_i = a$. Conversely, pad any subset-sum solution with zeros to exactly m members. This proves hardness for the linked subclass.

For common g , scan descending works and select i exactly when $(h + 1)g + w_i \leq T$, with h previously selected. Its j th selected position is no later than the j th position of any feasible subset: inductively that comparison job passes the same rank- j test. Greedy therefore selects at least as many jobs and at least as much work at every matched rank, maximizing saved work.

For the whole frontier, apply this greedy rule at thresholds $\{\Gamma\} \cup \{w_i + jg > \Gamma : 1 \leq i, j \leq n\}$, compute each witness's actual (A, D) and prune. Any subset C has threshold $T = \max(\Gamma, K(C))$ in this set. A maximum-work witness C' at T has $K(C') \leq T$ and $\Omega(C') \geq \Omega(C)$, so $A(C') \leq A(C)$ and $D(C') \leq D(C)$; here $D(C) \geq T$. At most $n^2 + 1$ thresholds and a linear scan each give the bound after sorting. $\square$

Protected cost *alone* remains polynomial on every DAG when $c_i = 0$. In Theorem 10, changing a fresh job to cached lowers all later fresh-body terms, and its new own term is bounded by the old no-write protected total. Thus all-cached is optimal for some order, giving $\min_{\pi} \max_i (\sum_{j \text{ through } i} g_j + w_i)$. This is Lawler's unchanged precedence algorithm [37], with processing g_i and tail w_i . The hardness concerns simultaneous caps with entry work in *both* coordinates; it does not contradict the body-only results.

7.2 Unit Entry Costs Make Precedence Hard

With $g_i = c_i = 0$, an optimal policy under $W \leq \Omega + M$, $M \geq 0$, can cache every job with $w_i \leq M$: in any fixed topological order, each such replacement introduces a mismatch term bounded by the old no-write protected load and decreases later fresh-body prefixes. Set $C = \{i : w_i \leq M\}$, $F = J \setminus C$, processing $p_i = w_i \mathbf{1}[i \in F]$ and tail $q_i = w_i \mathbf{1}[i \in C]$. Minimizing protection is exactly Lawler's objective $\max_i (C_i^P + q_i)$ [37], where C_i^P is processing completion time, including zero processing times. Its inherited rule at the at most $n + 1$ thresholds $M \in \{0\} \cup \{w_i\}$, followed by pruning, gives the complete frontier in $O(n^3)$ operations on any DAG. A cap below Ω is infeasible.

THEOREM 12 (UNIT-GUARD PRECEDENCE BOUNDARY). *With $c_i = 0$ and all $g_i = 1$, joint-cap decision is strongly NP-complete already on height-two DAGs with at most two predecessors per job and $w_i \in \{1, n + 1\}$. Nevertheless, every DAG with a common guard price g has at most $n^2 + 1$ distinct frontier points.*

PROOF. Membership follows from Theorem 10. We adapt Garey–Johnson's incidence/prefix construction [1, Theorem 2]. Take a simple graph and $k \geq 3$, with $K = \binom{k}{2}$; settle too few vertices/edges separately. Add an isolated edge and two vertices, preserving k -cliques and giving augmented counts $v \geq k + 2$, $m \geq K + 1$. Create v vertex jobs of work 1 and m edge jobs of work $H = n + 1$, $n = v + m$, each following its endpoints. Set

$$D = k + K, \quad M = D + H, \quad K_W = \Omega + M, \quad K_P = n + (m - K)H.$$

Then $n > D$ and $K_P > M > n$; the extra edge supplies room for mismatch cost.

Given a clique, order its vertices, its edges, other vertices, then other edges. Cache all vertices and the clique edges; finish other edges fresh. Every cached prefix has $F_i = 0$ and $\text{pos}(i) + w_i \leq M$. Theorem 10 gives $W \leq K_W$ and $P = K_P$.

Conversely, extract (π, C) from one fixed no-write execution of any feasible program. Its protected load $n + \Omega(F) \leq K_P$ forces at least K cached edge jobs. For each, a separate own-gate one-write witness on that same prefix requires $\text{pos}_{\pi}(i) + H \leq M$, hence position at most D . The first D jobs therefore contain at least K edges and at most k vertices, including every endpoint. A simple graph on at most k vertices with at least $\binom{k}{2}$ edges is a k -clique. Padding creates none. All numerical parameters are $O(n^2)$, giving strong hardness.

For common g , Theorem 10 restricts the first coordinate to

$$\{\Omega + ng\} \cup \{\Omega + w_i + jg : 1 \leq i, j \leq n\}.$$

Only the least second coordinate for each first coordinate can be nondominated, proving the cardinality bound. $\square$

Polynomial output size does not ensure polynomial search. The supplement preserves the tight $2^n - 1$ frontier construction for independent jobs with heterogeneous guards [26].

8 Formal Corroboration and Operational Evaluation

Inputs are fixed before execution; post-outcome reanalyses and all failures/corrections remain explicit and archived.

Author executions use an Apple M2/24 GiB, Python 3.10.9/3.12.14 and OpenJDK 17.0.19; the cited map version was inspected, not rebuilt. Hashes bind code, inputs and traces. Codex assisted the research and writing; Claude supplied proof/direction critiques and manuscript reviews. No human study or independent certification is claimed.

An input fixes graph/works/prices; a root adds budget/retries. Paths are complete branch sequences, executions native invocations, and pairs compare policies on one input. Coordinates add evaluation budgets. The studies overlap.

8.1 RQ1: Bounded Corroboration of the Proofs

Table 3. Bounded checks of the retained claims. Populations overlap; finite agreement does not prove all-program completeness.

Obligation	Evidence and comparison
General call/ L values	Direct minimax versus formulas: 265,629 agreeing roots from 5,421 DAG/weight inputs.
General retry frontier	Immediate and retained-preparation finite models: 96,795 agreeing cases; $n \leq 4$, works $\{1, 2, 4\}$, $r = 0, 1, 2$.
Polynomial formulas	Retrospective: 1,998 independent and 32,526 enough-retry DAG cases (720 overlap). Prospective: 13,850 agreeing roots and 540,230 independently interpreted paths.
Unknown-budget curves	647 fixed independent roots and 34,814 interpreted paths; exact least curves agree, zero mismatches. Earlier 713 general roots remain.
DAG/observation boundaries	36 scope roots, zero mismatches; four erasure roots, 8,672 trees and 102,122 paths. All 8,668 protection violations retained.
Charged representation	14,696 inputs; all order/mask and interpreted-path comparisons agree.
Free-comparison boundary	764 independent and 836 common-guard inputs; all policy/cap comparisons agree.
Unit-guard precedence	3,208 source cases; 1,399 reductions (728 yes/671 no); 675 common-guard DAG cases. All agree.
Changed call unit	13,176 supplied roots from 1,098 inputs and 222 unaware curves from 74 inputs; all agree with Proposition 6.

The prospective general-retry study uses 461 multisets, $n \leq 6$, works $\{1, 2, 4, 7, 11\}$, $r \leq 3$, $B \leq n+r$. All 9,240 first-excluded-budget controls violate L , preserving Q ; Boolean-value cases are archived.

Both wrappers cover colliding/separate and tree bins, and canonical/allocating outputs. All 6,112 supplied and 10,192 unaware pairs agree. The unaware selector sees no B , flag or diagnostics.

Twelve corruptions reject; four of 52 flag-ignoring controls violate L . All outcomes and tightness witnesses remain archived.

Subset-sum checks cover 420 targets (339 yes/81 no); seven alterations reject. Greedy reuses the 836 common-guard inputs. The unit-guard study enumerates all graph/ k cases on 3–5 vertices, retaining 1,809 trivial negatives. Constructive paths respect both caps; each policy’s separate maxima attain them. Fee recurrences translate complete frontiers; three alterations reject. Full masks, thresholds, controls and paths are archived. These checks corroborate witnesses, not asymptotic proofs.

8.2 RQ2: A Native Traversal and Its Guard Boundary

Linux v6.12 `dentry_path_raw` first walks mutable `dentry` parents/names under RCU, validates `rename_lock`’s sequence, and on failure rebuilds under its reader-exclusive lock [10]. Pathname lock contention motivated the upstream optimistic fallback [12]. Our KUnit-only variant prepares a second traversal with a *new even* sequence, then acquires the lock and validates that saved sequence: reuse on match, reset/rebuild on mismatch. Every traversed component must be initialized, with no concurrent initialization; a valid reference alone is insufficient. Initial `dentry`/name publication must precede parallel reading; subsequent relevant changes must participate in the global sequence, without wrap. This covers the initialized `d_exchange` fixtures, not every filesystem: `d_mark_tmpfile` changes names without that sequence during initialization [11]. Mount traversal and `getcwd` are excluded.

Assume a private valid buffer with $1 \leq \text{capacity} \leq \text{INT_MAX}$, nonnegative observed lengths fitting `int`, valid `dentry`/RCU lifetimes, and successful stable-name copies. Let $C = \text{capacity} - 1$ after NUL. In the original factored loop, attempted names v , requested copy bytes c , fault-fill bytes f and successful slashes s satisfy $v \leq s + 1$, $f \leq c$ and $c + s \leq C$. Thus body events $T = 1 + v + c + f + s \leq 2C + 2$. For first/second validation failures X, Y , fresh even capture gives $X + XY \leq \min(B, 2)$ for actual global writer sections, including unrelated renames. Count one Q unit per lockless validation or guarded validation/completion block, not per native API call. The candidate has $Q \leq 2$, $T_{\text{total}} \leq (1 + \min(B, 2))(2C + 2)$ and $L_T = 0$ for $B \leq 1$. This is a single-job upper refinement: private-buffer return does not instantiate the full publication/observation contract. Guard/RCU instructions, waiting and postambles are separate; the capacity-4096 ceiling 8192 is loose on short paths.

ARM64 Linux runs in two-CPU QEMU with lockdep and atomic-sleep checking. A pinned native `lock_rename/d_exchange` writer uses endpoint gates outside RCU and a nonsleeping mid-parent gate. The fixed matrix has 1,920 rows (948 paths/972 expected overflows) and four detected defects. A fixture oracle checks strings, offsets, canaries and post-return persistence against quiescent upstream. Outputs/body records agree without lock/RCU diagnostics. Even nonwrapping sequence brackets confirm designated writer counts and all prior output/body rows. Returning unvalidated overflow, failing to restore the buffer or ignoring final validation breaks output. Reusing the first sequence preserves output but violates the protection bound at $B = 1$. No copy-fault/allocation failure is injected; native-time performance is unmeasured.

All 384 cached/two-cheap pairs agree on output and body vectors; 138 first-invalid/second-valid pairs avoid the guard, while 144 two-failure pairs need another logical block. The cap comparison depends on its unit. One public `dentry_path_raw` invocation remains one under every policy. Counting six named primitives—RCU enter/exit, sequence begin/retry, and exclusive lock/unlock—gives $4R + 2G$, where R counts outside traversals. The cached/two-cheap counts are 4/4, 10/8, 10/10 for zero, one and two observed failures: both universal maxima are 10, despite logical caps 2/3. These are source call counts, excluding spinning and underlying instructions. No caller-required $Q = 2$ SLA was found. A stronger native name-snapshot comparator copies inline names or retains

Table 4. Source event counts at the scripted gates. Initial path is /driver/cache/entry/leaf; first exchange moves it to /alternate, second restores it. G counts reader-exclusive acquisitions.

Policy	One write				Two writes			
	T	L_T	Q	G	T	L_T	Q	G
Upstream	41	12	2	1	58	29	2	1
Cached	41	0	2	1	70	29	2	1
Two cheap, then locked	41	0	2	0	70	29	3	1

external-name references under protection, then materializes outside. Its allocation, dentry locks, capture steps and reccounts are separate from L_T ; five 64-slot overflows retain the discarded capture work before protected fallback. Its body-only zero is not zero total protected work or a lower bound for larger workspaces. All adverse results and the first 1,536-row matrix remain; the populations overlap.

8.3 RQ3: Preservation and Cost Limits in Roslyn

Roslyn’s `Workspace.cs` at 6c4a46a3 transforms an immutable solution outside a semaphore, validates identity inside and retries [13, 14]. Our `SourceText`-only patch switches after r shared failures to fresh (two-mode) or cached completion (three-mode). A stronger rebase reuses guarded document contents but still requires a protected merge. Lazy work, allocation, version generation and cache-dependent merges prevent fixed-work transfer. No upstream numerical cap or SLA was found.

All 236 authored cases agree; 12 corruptions reject. Full event/publication/snapshot records are archived. In 96 compiler processes, 2,556 project projections match declared expectations for references, parse options and linked text under cold/warm caches; eight corruptions reject. The original same-text deadlock and four missing-document timeouts remain archived; corrected patches pass 14 error cases and reentry controls. These selected checks use Roslyn on both sides. Custom loaders, changed-text reentry and general concurrent equivalence remain outside scope.

Arrival tests compare shared/per-update caps, two/three modes and both baselines. Eight balanced blocks serve 32 updates and 64 FIFO writers each at 0, 10, 100, 1000 μ s. All 2,560 measured and 512 warm-up units pass source/count checks; 12 corruptions reject. At $r = 2$, local/shared caps allow 96/34 calls, precluding equal-cap comparison.

Shared three/two foreground-time ratios at 0/10 μ s are 1.72 [1.40,2.11] and 1.58 [1.17,2.16], favoring two modes. Slower-arrival and all writer-response intervals include 1. Geometric means use eight paired block ratios and descriptive 95% intervals from 10,000 fixed block-bootstrap samples without multiplicity correction. Baseline ratios 0.83–1.08 all include 1; instrumentation overhead is unidentified. Authored arrivals, eight blocks and one host limit inference; the supplement retains all mixed results.

9 Related Work and Research Lineage

Fallback and contention. Protected reexecution [9, Section 3.3], bounded lock-elision/RPC retries [29, 30, 34, 35], irrevocable prefix preservation [36], contention management [7, 27, 28] and switching/tuning [8, 44, 45] are established. HTM aborts need not consume writes [31], and RPC can fail. Their progress/time objectives differ from persistent-job W/L caps.

Synthesis. Černý et al. [18] synthesize retries, guards and caches for weighted limit-average safety (PSPACE by erratum); later work optimizes protection/profile costs [19] or preemption-safe

locks [20]. We inherit global-budget/information-state minimax DP [15, 24, 32], backward induction and Pareto composition; our results supply all-program domination and exact polynomial regions.

Scheduling with testing. At $r = 0$, fresh completion is untested execution of cost $u_i = w_i$; cached completion tests for $t_i = w_i$, then executes for $p_i \in \{0, w_i\}$. Thus W counts testing plus execution, and L execution alone. A deterministic tree’s mismatch set can be fixed as hidden bad-job labels, and own-gate writes realize those labels. Dynamic writing alone does not distinguish this canonical game.

Dürr et al. [38] introduce adversarial testing; Buld and Schulz [41] and Albers and Eckl [42] allow unequal test times/upper bounds, for competitive time objectives. Damerius et al. [39] separate testing expenditure and execution with a *policy-side* budget. Our B limits adverse outcomes; preparations remain allowed at $B = 0$. Dufossé et al. [40] study minimax testing trees with durations revealed even for untested execution; polynomial adaptive optimality assumes their two-phase conjecture. Albers and Janke [43] bound failing jobs for irrevocable parallel-machine assignment. Our specialization returns causal completion policies for separate absolute W/L caps and retry slack. The informed comparator knows only B .

Precedence and reuse. Section 7 embeds inherited precedence [37], weighted-deadline [2] and incidence-hardness [1] results. Universal restart [46] concerns expected Las Vegas time. Build and incremental systems [5, 6, 33] may reuse partial work outside our whole-kernel interface.

10 Limitations and Threats to Validity

Section 2 requires deterministic, fully charged whole kernels and excludes retained guards. Least unaware work curves for general DAGs remain open.

Supplied B is a premise. Native costs are uncalibrated; instrumentation, caches and runtime compilation limit timings. The archive retains six scaling timeouts, retained-guard/simple-rule counterexamples, reversed disjoint-bin timings, scalar gains in only 2/96 cells and successful cost fits in 2/8. All-budget compilation can cost more than requested-value computation. In all 16 Valkey digest cells, cooperating writer maintenance beats reader repair in time/transfer, with mixed maximum command times. These larger interfaces delimit our claims. Finite/AI-assisted checks are neither mechanical proofs nor independent reproduction; human benefit is unmeasured.

11 Conclusion

For supplied $B = 1$, no retry slack and free comparisons, charging unit guard work in both coordinates changes DAG joint caps from polynomial to strongly NP-complete despite polynomial output size. Independent-job frontiers, call-count sensitivity and mixed source results delimit transfer.

Data Availability

Code, fixed inputs, all outcomes/corrections, proofs, replay commands and licenses are archived [26]. References [25, 26] are snapshots of the same artifact lineage; they are not independent novelty baselines.

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